MTH312

Timed Online Assignment – January Semester 2022

Real Analysis I

Tuesday, 17 May 2022 **1:00 pm – 3:00 pm**

Time allowed: 2 hours

INSTRUCTIONS TO STUDENTS:

1. This Timed Online Assignment (TOA) contains **EIGHT (8)** questions and comprises **FIVE (5)** pages (including cover page).

2. You must answer **ALL** questions.

3. If you have any queries about a question, or believe there is an error in the question, briefly explain your understanding and assumptions about that question before attempting it.

4. You MUST **submit your answers via Canvas (similar to TMA submission) at the end time of this TOA** (as stated on this cover page). The 15 minutes grace period as shown on Canvas is strictly meant for technical issues encountered during submission. Thereafter, you will not be able to submit your answers and you will be considered as having withdrawn from the course. **No appeal will be allowed.**

5. You are to include the following particulars in your submission: TOA Course Code, your Full Name, and SUSS PI Number. Name your submission file as TOACourseCode_FullName_StudentPI. Use underscore and NOT Space. *Example: TOAXYZ_RaphaelLee_T1923161 (omit D/O, S/O).*

6. Your submission should consist of only one file and **must not exceed 500MB** in size. The file must be a PDF or Microsoft Word file saved in .doc or .docx format. All answers are to be typed. Flowcharts and graphs may be scanned or photographed and embedded in the PDF or Word file provided it does not exceed the file size limit of 500MB.

7. For answers which cannot be typed and require to be hand-written, ensure that question number is clearly stated on each page. All uploaded hand-written answers must be clear, readable and complete. Marks will not be awarded for un-readable or incomplete images.

8. To prevent plagiarism and collusion, your submission will be reviewed by Turnitin. The Turnitin report will only be made available to the marker and you will not be able to view it.

9. The University takes plagiarism and collusion seriously, and your Turnitin report will be examined thoroughly as part of the marking process.

10. Marks will only be awarded if FULL working is shown.

Answer all questions. (Total 100 marks)

Question 1

Compute the limits of the following sequences.

(a) $x_n = \sqrt{4n^2 + 3n + 3} - 2n$

(5 marks)

(b) $y_n = \sum_{i=1}^{n} \frac{i}{3n^2+i}$

(6 marks)

(c) $z_n = \frac{2^n}{n!}$. Note that $n! = 1 \times 2 \times \ldots \times n$ is the factorial of $n!$

(6 marks)

Question 2

Give an example of two sequences $(x_n, n \in \mathbb{N})$ and $(y_n, n \in \mathbb{N})$ satisfying the following two conditions:

(a) $\lim_{n\to\infty} \frac{x_n}{y_n} = 1$ and

(b) $\lim_{n\to\infty} (x_n - y_n)$ does not exist.

(6 marks)

Question 3

Assume $(x_n, n \in \mathbb{N})$ is a decreasing sequence of real numbers such that $x_n > 0$ for all $n \in \mathbb{N}$. Let $s_n = \sum_{i=1}^{n} x_i$ be the associated series. If $\lim_{n\to\infty} s_n$ exists, prove that $\lim_{n\to\infty} nx_n = 0$.

(12 marks)

Question 4

Let $(x_n, n \in \mathbb{N})$ be a sequence of real numbers and define $X_n = \sum_{i=1}^{n} |x_{i+1} - x_i|$. Assume $X_n \leq M$ for any $n \in \mathbb{N}$, where $M > 0$ is a positive constant.

Prove:

(a) The sequence $(X_n, n \in \mathbb{N})$ is convergent.

(5 marks)

(b) Use the Cauchy Criterion to show that the sequence $(x_n, n \in \mathbb{N})$ is also convergent.

(8 marks)

Question 5

Assume $f(x)$ is a function on $\mathbb{R}$ and $\lim_{x \to 0} f(x) = L$, where L is a real number. Suppose $a > 0$ is a constant and we define $g(x) = f(ax)$. Apply epsilon-delta arguments to prove that $\lim_{x \to 0} g(x) = L$.

(10 marks)

Question 6

Determine whether the function $f(x) = \frac{1}{x^2+1}$ is uniformly continuous on $\mathbb{R}$. Give the reasons.

(12 marks)

Question 7

Let $f: [0,2] \to \mathbb{R}$ be a continuous function such that $f(0) = f(2)$. Show that there exists $x_1,\ x_2 \in [0,2]$ satisfying the following two conditions:

(a) $|x_1 - x_2| = 1$ and

(b) $f(x_1) = f(x_2)$.

(15 marks)

Question 8

Let f and g be two continuous functions defined on $\mathbb{R}$. Define the maximum function h of f and g by $h(x) = \max\{f(x), g(x)\}$ for all $x \in \mathbb{R}$. Show that $h(x)$ is also continuous on $\mathbb{R}$.

(15 marks)

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